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Geosteering Thin Carbonate Reservoirs by the Near-Bit Gamma Ray Image (NBGR-Image)

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Abstract

Geosteering in thin carbonate reservoirs poses a significant challenge due to complex sedimentation, structural heterogeneity, and the presence of tight non-reservoir boundaries such as anhydrite and shale. Conventional at-bit resistivity tools often struggle to detect these transitions in real-time, especially in thin pay zones, leading to premature exits or poor reservoir contact. This paper presents the application of Near-Bit Gamma Ray Imaging (NBGR) as an effective geosteering solution for horizontal drilling in such environments.

This study focuses on a thin carbonate reservoir in the Middle East—a thin, highly heterogeneous carbonate unit bounded by anhydrite layers. Using a near-bit azimuthal gamma ray imaging tool, a case well was drilled through this zone, demonstrating enhanced boundary detection, real-time formation characterization, and improved well trajectory control. The tool's quadrant-based and sector-based imaging enabled proactive trajectory adjustments, especially in structurally complex areas exhibiting dip changes, truncations, and faults.

Results from the case study show that NBGR significantly improved reservoir contact and total productive footage by providing high-resolution gamma ray images just inches behind the bit. In contrast to resistivity-based methods, NBGR responded quickly to lithological changes and minimized delay in detection at reservoir boundaries. Integration with uphole density and resistivity tools provided quality control and helped confirm geosteering decisions in real-time.

While NBGR offers simplicity, cost-effectiveness, and operational ease, limitations were observed in internal reservoir facies interpretation. To address this, enhancements such as increasing sector resolution (e.g., from four to eight or more sectors) and combining NBGR with azimuthal density or resistivity imaging are recommended.

This paper introduces NBGR as a practical and innovative alternative to conventional imaging tools in thin carbonate reservoirs. It offers valuable insight into real-time geosteering optimization, ultimately improving well placement, reducing non-productive time, and supporting more effective field development strategies in complex geological settings.

Introduction

Geosteering horizontal wells in thin carbonate reservoirs presents significant operational challenges due to complex sedimentary structures, heterogeneity in porosity and permeability, and the presence of tight, high-resistivity non-reservoir boundaries such as anhydrite and shale. These complexities often lead to premature exits from the target pay zone, compromising reservoir contact and well productivity.

The carbonate reservoir, particularly the target interval, represents a prolific yet technically demanding carbonate unit. Characterized by shoaling-upward depositional cycles and bounded by anhydrite, this zone typically exhibits a true vertical thickness of less than 5 feet, making precise geosteering essential. Traditional resistivity-based LWD tools frequently struggle in these conditions due to limited sensitivity to lithological changes between similar resistivity units and delayed response at formation boundaries.

This paper explores the application of Near-Bit Gamma Ray Imaging (NBGR) as an effective geosteering alternative in thin, high-resistivity carbonate reservoirs. By providing real-time, high-resolution gamma ray images just inches behind the bit, NBGR enhances boundary detection, improves trajectory control, and optimizes reservoir contact. The study presents a field case from a thin carbonate reservoir, evaluating NBGR's operational performance and comparing its effectiveness against conventional resistivity-based tools in a complex structural setting.

Statement of Theory and Definitions

The reservoir is characterized by shoaling-upward depositional cycles, typically consisting of lower carbonate units overlain by evaporite sequences. Carbonate depositional cycles are commonly divided into discrete facies sequences that reflect shifts in depositional energy and sediment supply. Within these cycles, vertical variations in porosity and lithology—driven by changes in packing, grain size, and diagenetic overprint—create thin, high-quality layers that are challenging to detect with conventional geosteering tools (Choquette and Pray, 1970; Lucia, 2002).

Structurally, the region is influenced by a regional graben system, a series of fault systems and lineaments extending up to 90 km in length, forming an east-concave arc approximately 500 km long (Weijermars, 1998). Faulting in this region began in the Late Cretaceous and continued into the Eocene, introducing significant dip variations, structural traps, and lateral facies terminations (Ibrahim et al., 2012). These structural complexities complicate well placement, requiring real-time, high-resolution subsurface imaging to accurately steer horizontal wellbores within the thin reservoir interval.

The primary challenge in thin carbonate geosteering lies in accurately detecting and reacting to formation boundaries in real time. Traditional resistivity-based LWD tools often fall short in this regard due to delayed response time and reduced sensitivity to lithological changes between similarly resistive units (e.g., reservoir vs. anhydrite). Therefore, alternative techniques, such as Near-Bit Gamma Ray Imaging (NBGR), are required to maintain well trajectory within the productive zone and avoid unintended exits into non-reservoir intervals.

Description and Application of Equipment and Processes

To address the challenge of steering within a thin, heterogeneous carbonate reservoir, a Near-Bit Gamma Ray Imaging (NBGR) tool was deployed in a horizontal section targeting a thin carbonate reservoir. This tool was designed specifically for geosteering applications, offering real-time azimuthal gamma ray measurements directly at the bit. It enables immediate detection of lithological boundaries with enhanced resolution compared to conventional gamma ray sensors placed farther uphole in the Bottom Hole Assembly (BHA).

The NBGR tool operates by detecting natural gamma ray emissions radially around the borehole wall. During sliding mode, the tool records data in four quadrants. While rotating, the tool bins gamma ray data into multiple sectors (e.g., octants or hexants), thereby constructing a near-bit borehole image. These images are transmitted to the surface in real-time, allowing geosteering engineers to interpret bed boundaries, structural dip, and trajectory deviations quickly and accurately.

The Borehole Image Unwrapping represents the borehole wall as a 2D surface, where the center of the image corresponds to the bottom of the wellbore, and the edges correspond to the top. Curvature in the image (e.g., sinusoidal patterns) indicates whether the trajectory is cutting upwards or downwards through dipping beds. This feature provides proactive warning of potential exits and informs inclination changes needed to stay within the reservoir, as shown in Figure 1.

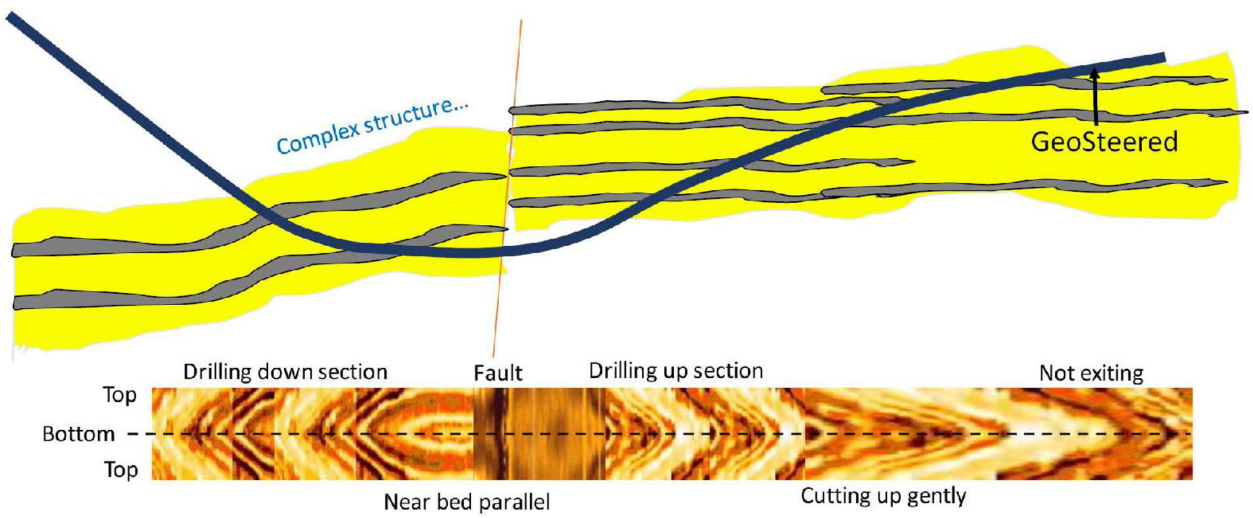

Figure 1—Real-time GR image indicates cutting relationship between trajectory and formation (Ma Jia Zheng, 2019).

To validate NBGR data and enhance interpretive confidence, the tool was run in conjunction with density and resistivity image tools positioned at various offsets up the BHA. This redundancy allowed quality control (QC) by comparing near-bit measurements with those obtained further uphole, shown in Figure 2.

This tool configuration was particularly effective when the well encountered structural features such as dip changes, minor faults, or formation truncations. In one instance, an abrupt image curvature—suggestive of an upward exit into the overlying anhydrite—was promptly identified. This allowed the geosteering geologist to reduce inclination and realign the trajectory with the reservoir,

In summary, the integration of NBGR imaging with auxiliary LWD tools provided a robust, real-time system that enabled high-precision geosteering in a thin, structurally complex carbonate reservoir. This approach minimized reservoir exits and maximized total reservoir contact.

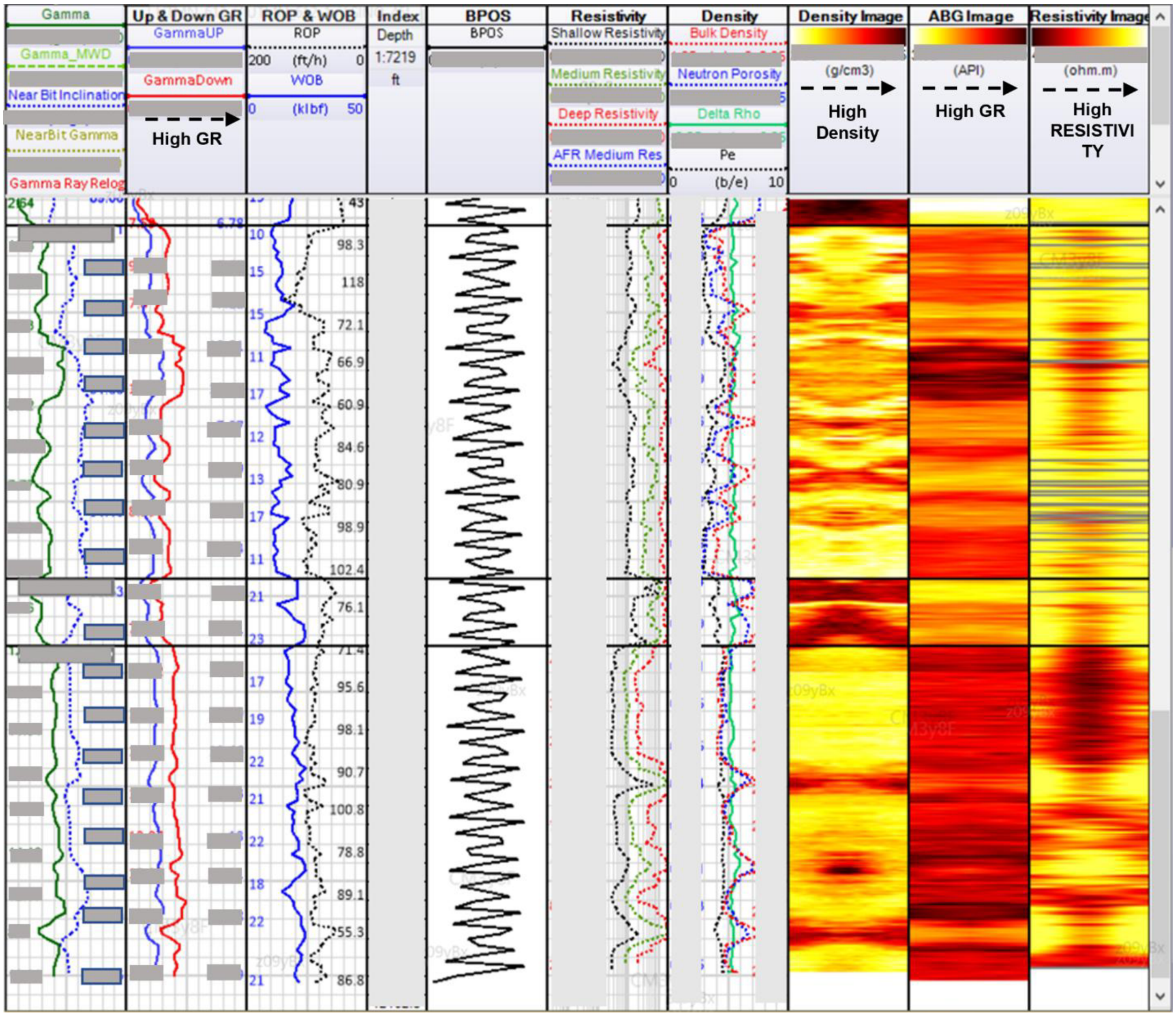

Figure 2—Real-time data while geosteering the well showing GR image offset relative to QC density and resistivity measurements.

Presentation of Data and Results

A horizontal well—designated KK—was drilled in a thin carbonate reservoir carbonate reservoir to test the application of Near-Bit Gamma Ray Imaging (NBGR) in a structurally complex, thin-reservoir environment. The target interval presented a true vertical thickness (TVT) of approximately 3 feet, with reservoir rock bounded by anhydrite above and below. This combination of minimal reservoir thickness and high-resistivity bounding facies presented a substantial challenge for conventional resistivity-based geosteering tools. as shown in Figure 3.

Prior to drilling, reservoir characterization showed that resistivity contrast across the anhydrite-reservoir interface was insufficient to generate a reliable geosteering signal. This would likely result in delayed detection of boundary crossings, increasing the risk of premature exit from the pay zone. Based on these findings, the geosteering team selected the NBGR tool as the primary LWD tool.

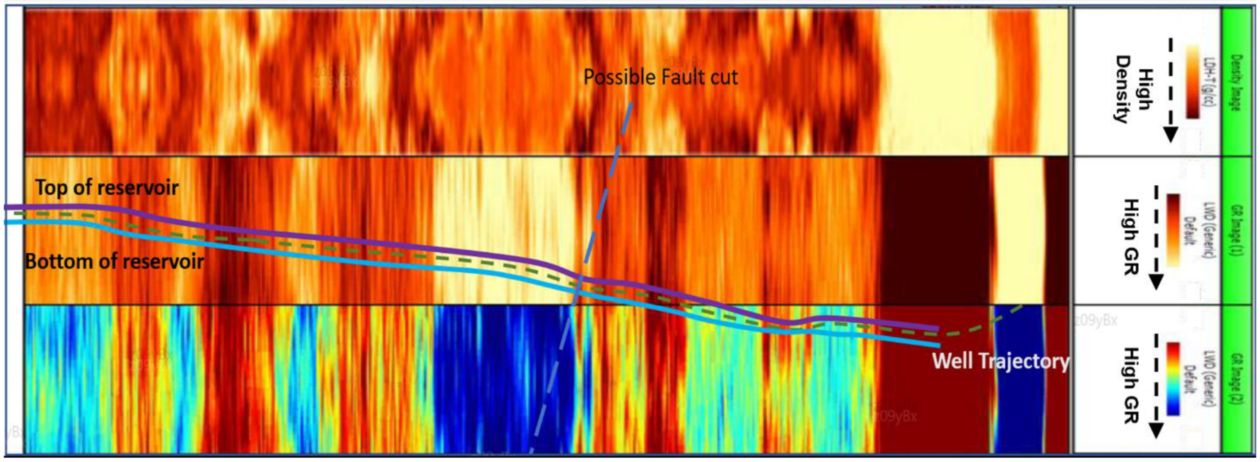

Figure 3—Conceptual cross section showing the well trajectory and the geosteering scenario.

After successful landing at the reservoir's structural top, the well was drilled horizontally at a controlled inclination. In the early stages, NBGR imaging revealed a distinct sinusoidal curvature—commonly referred to as a "happy face"—which indicated that the well was beginning to cut upwards toward the overlying anhydrite. This real-time visual cue enabled immediate corrective action: the inclination was reduced, and the trajectory was adjusted to remain within the reservoir.

As drilling progressed, the NBGR tool detected several instances of dip changes and potential truncations, suggesting the presence of minor faults or structural drag along the trajectory. These were later confirmed by azimuthal density and resistivity image logs uphole in the BHA. The integrated tool string, which provided redundancy and cross-validation, allowed the team to make timely corrections and maintain productive trajectory alignment.

Despite multiple dip changes and a complex structural setting, the well maintained consistent placement within the reservoir. The NBGR tool's high-resolution imagery enabled early detection of formation changes, proactive adjustments, and improved overall steering efficiency. Compared to traditional at-bit resistivity tools, NBGR offered more timely formation response, particularly in high-resistivity and low-contrast lithologies.

Additionally, offset image log comparisons with density and resistivity confirmed the accuracy of NBGR observations, validating its use as a standalone geosteering decision-making tool in thin reservoirs.

Conclusions

The application of Near-Bit Gamma Ray Imaging (NBGR) in the geosteering of thin carbonate reservoirs has proven to be a reliable and responsive alternative to traditional resistivity-based imaging tools. In the case of a thin carbonate reservoir, where the pay interval is less than 3 feet TVT and bounded by high-resistivity anhydrite layers, NBGR enabled superior boundary detection, improved trajectory control, and optimized reservoir contact.

Key conclusions from this study include:

- Real-time imaging at the bit allowed proactive well path adjustments based on formation responses, minimizing unintended exits from the reservoir.
- NBGR demonstrated faster response to lithological changes compared to at-bit resistivity tools, which often underperform in high-resistivity environments.
- Integration with azimuthal density and resistivity image logs provided quality control, enhancing confidence in the geosteering decisions.

- The use of NBGR significantly increased net pay exposure and reduced non-productive time, particularly in structurally complex zones.

These results confirm that NBGR is an effective geosteering solution in environments where traditional methods are limited by formation properties, tool lag, or resolution constraints.

Novel/Additive Information

This paper introduces the application of Near-Bit Gamma Ray Imaging (NBGR) as a cost-effective and technically superior alternative to traditional at-bit resistivity tools for geosteering in thin carbonate reservoirs. The key innovations and contributions include:

- Demonstrating NBGR performance in tight, high-resistivity boundary settings, specifically in a thin carbonate reservoir.
- Providing a detailed methodology for interpreting NBGR images in real-time, including sinusoidal curvature analysis for dip determination.
- Highlighting the importance of integrated tool string design, pairing near-bit and offset imagers for enhanced geosteering accuracy and data confidence.
- Suggesting future enhancements such as increasing sector resolution (from 4 to 8 or 16) and combining NBGR with other imaging modalities to further improve steering performance.

This work advances the operational understanding of NBGR and its potential in complex carbonate environments, contributing to more effective field development strategies.

References

Choquette, P. W., and Pray, L. C. 1970. Geologic Nomenclature and Classification of Porosity in Sedimentary Carbonates. *AAPG Bulletin* **54** (2): 207-250.

Craft, B. C., and Hawkins, M. 1991. *Applied Petroleum Reservoir Engineering*, second edition. Englewood Cliffs, New Jersey: Prentice-Hall.

Ibrahim, E., Kassem, O., and Al Bassam, A. A. 2012. Aeromagnetic data interpretation to locate buried faults in Riyadh Region, the Middle East. *Scientific Research and Essays* **7** (22): 2022-2030. https://doi.org/10.5897/SRE11.2243

Lucia, F. J. 2002. *Carbonate Reservoir Characterization: An Integrated Approach*. Springer-Verlag, Berlin.

Ma, J., Liu, X., Zhao, J. P., Qiu, X. X., Fang, J., Wang, X. F., Zhao, J. K., and Geng, G. 2019. RSS with Near-Bit GR Imaging Assist Accurate Steer Drilling in Shale Gas Project in Sichuan: A Fit-for-Purpose Solution for Shale Gas Drilling Case Study. *Schlumberger Technology Services*.

Pitcher, J. L., Schafer, D. B., and Botterell, P. 2009. A New Azimuthal Gamma Ray at-Bit Imaging Tool for Geosteering Thin Reservoirs. Presented at the SPE/IADC Drilling Conference and Exhibition, Amsterdam, 17-19 March. SPE-119421-MS. https://doi.org/10.2118/119421-MS

Powers, R. W. 1968. Lexique Stratigraphique International, Volume III, Asie, Fascicule 10b1, Arabia Saoudite. *Centre Nationale de la Recherche Scientifique*, Paris, 177 p.

Qin, Z., Tang, B., Wu, D., and Luo, S. 2020. A qualitative characteristic scheme and fast distance prediction method of multi-probe azimuthal gamma-ray logging in geosteering. *Petroleum Exploration and Development* **47** (6): 1188-1195. https://doi.org/10.1016/S1876-3804(20)60121-6

Weijermars, R. 1998. Plio-Quaternary movement of the East regional Block. *GeoArabia* **3** (4): 429-456.